stationary_combustion_r4.R

June 20, 2024

Contents

`Stationary_combustion` *Create gridded stationary combustion methane emissions maps*

Description

‘Stationary_combustion‘ writes up to 56 netcdf files of gridded methane emissions from stationary combustion sources, as well as optional visuals

Usage

```
Stationary_combustion(
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  EIA_API_key,
  plot_directory,
  State_Tigerlines,
  focus_city_tigerlines
)
```

Arguments

`NEI_file`
: Character providing the full filepath to the NEI county level CO data for the states within the domain. This data is available at https://www.epa.gov/air-emissions-inventories/2017-national-emissions-inventory-nei-data. At the bottom of the page there is a data query - to download the desired file the national/state/county value should be set to county, the geographic aggregation should be set to the states of interest (hold control and click to select multiple), and the CAP should be set to carbon monoxide. The resulting excel file has columns for different variables and rows for different sector - county - fuel combinations. The variables are State, State FIPS, County, Sector, County FIPS, Pollutant, Pollutant Type, Emissions, and Unit of Measure, though the county, pollutant, pollutant type, and unit of measure are unused. There is an example file in the package's datasets folder that has been successfully used in this code available for reference.

`domain`
: SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system

`state_name_list`
: Character vector listing all states within the desired domain

`output_directory`
: Character providing the full filepath to save processed data

`inventory_year`
: Character indicating the desired year of data to use.

`verbose`
: Logical indicating whether to save additional output. This includes plots of the gridded methane emissions for each fuel-sector-inventory-variation combination as well as 2 summed plots for each inventory-variation combination - one for wood and one for all other sectors.

`County_Tigerlines`
: SpatVector. United States Census Bureau county shapefile downloaded in Main.

`Use_ACES`
: Logical indicating whether or not to use ACES to disaggregate from county-level to pixel level emissions. Either ACES or Vulcan must be used, though both can be.

`Use_Vulcan`
: Logical indicating whether or not to use Vulcan to disaggregate from county-level to pixel level emissions. Either ACES or Vulcan must be used, though both can be.

`ACES_directory`
: Character providing the full path to a folder containing the ACES sectoral CO2 inventories. Must include the residential, commercial, electric (elec), and industrial sectors. ACES v2.0 is available at https://doi.org/10.3334/ORNLDAAC/1943, though the hourly file should be averaged across hours to create an annually averaged inventory. Code to do this on a linux-based HPC system is available as the script "Annualize_ACES_seawulf.R" and the accompanying batch script "Annualize_ACES.sh". The year closest to "inventory_year" is used, but those further from that year are considered if the closest is unavailable.

`vulcan_directory`
: Character providing the full path to a folder containing the Vulcan sectoral CO2 inventories. Must include the residential, commercial, electric (elec_prod), and industrial sectors. Vulcan v3.0 is available at https://doi.org/10.3334/ORNLDAAC/1741, and the annual mean files should be used. The year closest to "inventory_year" is used. As all years are contained in the same file, it does not search for other years.

`ACES_year`
: Numeric providing the year of ACES data to use

`vulcan_band` Numeric providing the band of Vulcan data to use (1-6 = 2010 - 2015)

`stationary_combustion_by_state`
: Logical. Pulled from config file. indicating whether state-toal emissions should be distributed to the county scale as is. Either bystate or bydomain must be used, though both can be.

`stationary_combustion_by_domain`
: Logical. Pulled from config file. indicating whether state-toal emissions should be aggregated to the domain and then distributed to the county scale. Either bystate or bydomain must be used, though both can be.

`stationary_combustion_GHGI_data`
: Data frame. Pulled from config file. 1 by 15 data frame with consumption for each sector-fuel combination from the GHGI. Although the data is national, there is a state entry set to US_EPA. Consumption is in thousands of British Thermal Units (BTU). The GHGI is available at https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks and the values can be found in the Annexes in a table titled "Fuel Consumption by Stationary Combustion for Calculating CH4 and N2O Emissions (TBtu)". Includes every combination of residential (res), commercial (com), electric (elec), and residential (res) sectors with coal, petroleum (petr), natural gas (gas), and wood fuels with names using the abbreviations in parentheses (e.g., com_coal, elec_petr). Excludes res_coal and res_gas as residential coal does not exist in the U.S. anymore and residential gas is accounted for elsewhere.

`stationary_combustion_emission_factors`
: Data frame. Pulled from config file. 1 by 14 data frame with emission factors for each sector-fuel combination from the IPCC. Built equivalently to the GHGI_data, but without an entry for the state. Emission factors are in g/GJ, equivalent to kg/TJ. Default emission factors are available in IPCC 2006 volume 2: Energy, tables 2.2 through 2.5 https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html. The natural gas electric sector emission factor is instead pulled from Hajny et al., 2019 https://doi.org/10.1021/acs.est.9b01875. This is 5.7 kg/TJ, within uncertainties of the GHGI value of 4.1 kg/TJ, both of which are larger than the IPCC default of 1 kg/TJ.

`EIA_API_key` Character. Pulled from config file. API key to access SEDS data API. The API is described at https://www.eia.gov/opendata/ and one can register for a key with a link on the right hand side of this page.

`plot_directory` Character providing the full filepath to save figures. Only relevant if verbose = TRUE.

`State_Tigerlines`
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if verbose=TRUE.

`focus_city_tigerlines`
: SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html. Only relevant if a focus city was set in main and verbose=TRUE.

Details

This function calculates and grids methane emissions from stationary combustion. It uses the Energy Information Administration's (EIA) State Energy Data System (SEDS), Environmental Protection Agency's (EPA) Greenhouse Gas Inventory (GHGI), EPA National Emissions Inventory (NEI) and either the Vulcan or Anthropogenic Carbon Emission System (ACES) CO2 inventory. First, the ratio of SEDS national consumption by fuel-sector combination to the equivalent GHGI data is applied to SEDS state data so that they are approximately consistent with the GHGI. The state total consumption is then converted to emissions using emission factors. There are two variations at this step.

bystate: The state level emissions are then distributed to the county level using NEI CO concentrations to calculate weighting values for each county (i.e., county CO / state total CO).

bydomain: The state level emissions are summed to get a total for all states in the domain. This is distributed to the county level using NEI CO concentrations to calculate weighting values for each county (i.e., county CO / domain total CO).

Lastly, county total emissions are distributed using the ACES or Vulcan CO2 inventory.

This entire process is done separately for each fuel and sector using the GHGI, SEDS, and NEI data that are broken down by sector and fuel and CO2 inventories that are broken down by sector. Sectors considered are residential, commercial, industrial, and electric and the fuels considered are coal, natural gas, petroleum, and wood. Residential coal is ignored as it does not exist in the U.S. anymore and residential gas is ignored as it is accounted for elsewhere.

The necessary SEDS data will be automatically downloaded.

The GHGI is intended to capture all national emissions. The SEDS consumption data provides state-level consumption broken down by fuel (coal, natural gas, petroleum, wood, geothermal, solar, and electricity) and sector (residential, commercial, industrial, electric, transportation). The NEI data being used provides CO emissions at the county level. The GHGI is available starting in 1990 and is generally about 2 yearrs behind present day. SEDS data is available starting in 1960 and generally is about 2 years behind present day. NEI data is available beginning in at least 1990, is released every three years, and generally takes three years to complete (i.e., 2023 NEI is released in 2026). All data is annual. The SEDS data is at the state scale, NEI data is at the county scale, and GHGI data is at the national scale.

The GHGI is available at `https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-` The SEDS is available at `https://www.eia.gov/state/seds/seds-data-complete.php?sid=US` The NEI is available at `https://www.epa.gov/air-emissions-inventories/national-emissions-inventory-` ACES is available at `https://doi.org/10.3334/ORNLDAAC/1943` and Vulcan is available at `https://doi.org/10.3334/ORNLDAAC/1741`.

Each fuel-sector-inventory-variation combination is saved separately.

See references Vulcan and ACES

Value

Nothing is returned from the function, but the main outputs are up to 56 netcdf files of the methane emissions from stationary combustion. They are titled as "stat_comb_sector_fuel_variation_inventory.nc" where sector is abbreviated as com (commercial), res (residential), elec (electric), and ind (industrial); fuel is abbreviated as wood, petr (petroleum), gas (natural gas), and coal; and variation is bystate or bydomain.

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Examples

```
library(terra)
user_key = "__user_EIA_API_key__"
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)
Urban_Tigerlines <- vect("~/../Desktop/Urban_Tigerlines/tl_2018_us_uac10.shp")
focus_city <- terra::subset(Urban_Tigerlines,Urban_Tigerlines$NAME10 %in% "Philadelphia, PA--NJ--DE--MD")
Stationary_combustion(NEI_file="~/../Desktop/NEI_2017.xlsx",
                      domain=grid,
                      state_name_list=c("DE","MD","NJ","NY","PA"),
                      output_directory="~/../Desktop/",
                      inventory_year=2018,
                      verbose=TRUE,
                      Use_ACES=TRUE,
                      Use_Vulcan=TRUE,
                      ACES_directory="~/../Desktop/Inventories/ACES_v2.0",
                      vulcan_directory="~/../Desktop/Inventories/Vulcan_v3.0",
                      ACES_year=2017,
                      vulcan_band=6,
                      stationary_combustion_by_state=TRUE,
                      stationary_combustion_by_domain=TRUE,
                      stationary_combustion_GHGI_data=data.frame(
                        "State"="US_EPA",
                        "com_coal"=17,
                        "ind_coal"=517,
                        "elec_coal"=10554,
                        "res_petr"=975,
                        "com_petr"=801,
                        "ind_petr"=2062,
                        "elec_petr"=42,
                        "com_gas"=3647,
                        "ind_gas"=9484,
                        "elec_gas"=11553,
                        "res_wood"=544,
                        "com_wood"=84,
                        "ind_wood"=1407,
                        "elec_wood"=68),
                      stationary_combustion_emission_factors=data.frame(
                        "com_coal"=10,
                        "ind_coal"=10,
                        "elec_coal"=1,
                        "res_petr"=10,
                        "com_petr"=10,
                        "ind_petr"=3,
                        "elec_petr"=3,
                        "com_gas"=5,
                        "ind_gas"=1,
                "elec_gas"=5.4/(1.0550559*0.9), #g/mmbtu to g/GJ and low to high heating value (0.9)
                        "res_wood"=300,
                        "com_wood"=300,
                        "ind_wood"=30,
```

```
    "elec_wood"=30),
  EIA_API_key=user_key,
State_Tigerlines=vect("~/../Desktop/State_Tigerlines/tl_2018_us_state.shp"),
County_Tigerlines=vect("~/../Desktop/County_Tigerlines/tl_2018_us_county.shp"),
  focus_city_tigerlines=focus_city,
  plot_directory="~/../Desktop/plots/")
```

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